



Assure: Home Care Assistant

Wavve: Contactless Sleep Sensor

Wavve Hub & Kit: Elderly Home Care System

Your-cloud-restful-api

V1.8

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HISTORY

Date	Version	Modification
04-06-2023	V1.0	Release.
05-29-2023	V1.1	Sub-device connect/disconnect.
08-22-2023	V1.2	Radar connect
11-1-2023	V1.3	To push the firmware upgrade Status
11-26-2024	V1.4	Time format that complies with ISO-8601 ; Fall-alert-cancellation
02-24-2025	V1.5	2.9 Battery life of the sub-device; 2.10 Multi-target detection and target coordinates.
08-25-2025	V1.6	Update 2.8 Fall alert cancellation; Added 2.11 Room interference environment threshold learning completion
09-08-2025	V1.7	Update 2.4 Radar heart rate push; update 2.5.Add-on (sub-device) connect push Add 2.12 Door sensor status real-time push
03-03-2026	V1.8	Updated the request parameter for alarm push (2.1).

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1. Introduction

The RESTful APIs of your Cloud that are requested are listed in this document. Once the radar is connected to our Cloud, these interfaces will be used to communicate between the clouds so that the notifications can be sent to your Cloud. Once the APIs are requested by our Cloud, a reply of 200 from your Cloud is expected indicating that the connection is successful. Otherwise the connection is considered failed. If it fails, another two attempts will be made. If none were successful, this request will be abandoned. Please note that you will also be requested to send us the Base url of the HTTP service of your RESTful, for example (<http://xxx.xxx.com:80>).

2. Cloud-to-Cloud RESTful APIs

2.1. Alarm push

Url: /api/alarm

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Description: To receive alarm push notify. All the alerts or alarms will be pushed via this interface. For example, FallDetect(0x0009)

Example:

```
{
  "alarmEndTime": "",
  "alarmStartTime": "",
  "alarmZonedEndTime": "",
  "alarmZonedStartTime": "",
  "alarmType": 0,
  "deviceMac": "",
  "deviceName": "",
  "functionId": 0,
  "radarId": "",
  "alarmData": null
}
```

Request parameter:

Parameter	Description	In	Required ?	Data Type	schema
alarm dto	Alarm dto	body	true	Alarm dto	Alarm

Parameter	Description	In	Required ?	Data Type	schema
					dto
alarmEndTime	alarmEndTime(yyyy-MM-dd HH:mm:ss)		true	string(date-time)	
alarmStartTime	alarmStartTime(yyyy-MM-dd HH:mm:ss)		true	string(date-time)	
alarmZonedStartTime	ISO-8601 standard datetime		true	string(date-time)	
alarmZonedEndTime	ISO-8601 standard datetime		true	string(date-time)	
alarmType	alarm type , 1:remind 2:alert		true	integer(int32)	
deviceMac	Sub device MAC address		false	string	
deviceName	Sub device name		false	string	
functionId	Please refer to the alert type enumeration value list for specific values		true	integer(int32)	
radarId	Radar id		true	string	
alarmData	Alarm extension data. If functionId is 5 (Fall alert), it contains the fall point coordinates xy values		false	object	

Response status:

Status code	Description	Schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Example:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.2. Radar connect push

Url: /api/radar/connect

Method: POST

Produces: application/x-www-form-urlencoded, application/json

Consumes: */*

Note: To receive the radar connect push notify. It is an SDK 0x01 function.

Example:

```
{
  "firmwareVersion": "",
  "online": false,
  "radarAddress": "",
  "radarId": "",
  "radarType": 0,
  "serverTime": "",
  "zonedServerTime": "",
  "assignedRoomId": ""
}
```

Request parameter:

Parameter	Description	In	Required?	Data Type	schema
radar information dto	Radar information dto	body	true	Radar information dto	Radar information dto
firmwareVersion	radar firmware version		true	string	
online	radar is online		true	boolean	
radarAddress	radar connect ip address		false	string	

Parameter	Description	In	Required?	Data Type	schema
radarId	radar id		true	string	
radarType	radar type		true	integer(int32)	
serverTime	current server date time		ture	string(date-time)	
zonedServerTime	Time in the format of ISO-8601		true	string(date-time)	
assignedRoomId	ID of the room that the radar is assigned to.		false	integer(int64)	

Response status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Example:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.3. Radar disconnect push

Url: /api/radar/disconnect

Method: POST

Produces: application/x-www-form-urlencoded,application/json

Consumes: */*

Note: To receive radar disconnect push notify.

Example:

```
{
  "firmwareVersion": "",
  "online": false,
  "radarAddress": "",
  "radarId": "",
  "radarType": 0,
  "serverTime": "",
  "zonedServerTime": ""
}
```

Request parameter:

Parameter	Description	In	Required?	Data Type	schema
radar information dto	Radar information dto	body	true	Radar information dto	Radar information dto
firmwareVersion	radar firmware version		true	string	
online	radar is online		true	boolean	
radarAddress	radar connect ip address		false	string	
radarId	radar id		true	string	
radarType	radar type		true	integer(int32)	
serverTime	current server date time(yyyy-MM-dd HH:mm:ss)		true	string(date-time)	
zonedServerTime	ISO-8601 standard datetime		true	string(date-time)	

Response status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Example:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.4. Radar heart rate push**Url:** /api/radar/heart-rate**Method:** POST**Produces:** application/x-www-form-urlencoded, application/json**Consumes:** */***Note:** To receive radar heart rate push notify. It is a Wavve SDK 0x03e8 function.

Please note that the confidence of breath and heart rate data is only available to firmware version 2.4.21.31 and later. It is recommended to use heart rate with confidence $\geq 90\%$, and breath data higher than 80%. Please note that although the confidence figure for the heart rate data is standardized probability between 0 to 1, the confidence for the breath is actually the signal over noise ratio and sometimes can be over 1.

Example:

```
{
  "heartRateValue": 0,
  "personExist": false,
  "radarId": "",
  "respiratoryValue": 0,
  "heartConfidence": 0.0,
  "breathConfidence": 0.0
}
```

Request parameter:

Parameter	Description	In	Required?	Data Type	schema
heart rate data dto	Heart rate data dto	body	true	Heart rate data dto	Heart rate data dto
heartRateValue	heart rate value		true	number(float)	

Parameter	Description	In	Required?	Data Type	schema
personExist	Moving target detected within the FOV		true	boolean	
radarId	radar id		true	string	
respiratoryValue	respiratory value		true	number(float)	
heartConfidence	Heart rate confidence level		false	number(float)	
breathConfidence	Respiratory confidence level		false	number(float)	

Response status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Example:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.5. Add-on (sub-device) connect push

Url:/api/sub-device/connect

Method: POST

Produces: application/x-www-form-urlencoded,application/json

Consumes:*/*

Note: To receive the push notification of the connection status of the add-ons.

Example:

```
{
  "deviceMac": "",
  "deviceName": "",
  "deviceType": 0,
  "onlineState": false,
  "radarId": "",
  "roomId": 0,
  "firstAdded": false
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
subDeviceDTO	SubDeviceDTO	body	true	SubDeviceDTO	SubDeviceDTO
deviceMac	Sub device MAC address		false	string	
deviceName	Sub device name		false	string	
deviceType	Sub device type. Please refer to the enumeration value list for specific values		false	integer(int32)	
onlineState	Sub device online state. true: online false: offline		false	boolean	
radarId	radar id		true	string	
roomId	Assigned room id		false	integer(int64)	
firstAdded	Identifies sub-device binding status: • true: First time bound to the hub • false: not the first time bound.		true	boolean	

firstAdded field detailed description

● **First binding (true) definition:**

When the Zigbee network is connected for the first time to the hub and records of this device doesn't exist in the database.:

- 1) **New binding:** the Zigbee network of the hub is enabled via API and the sub-device is connected to the hub for the first time.
- 2) **Binding after a reset:** the records of the sub-device is deleted from the database and this sub-device is later registered to the same hub or a new one.
- 3) **Binding to a different hub:** the records of the sub-device are not deleted from the database and the sub-device is directly bound to a new hub.

● **Non-First binding (false) definition:**

All other scenarios when the sub-device is automatically reconnected to the hub other than the ones described above. For example:

- 1) **Reconnection caused by the hub:** the hub goes offline and then back online and the Zigbee network is restored.
- 2) **Reconnection caused by the sub-device:** the sub-device goes offline and then reconnects to the hub.
- 3) **Network recovery:** any automatic reconnection after an abnormal disconnection of the Zigbee network.

Response status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.6. Add-on (sub-device) disconnect push

Url: /api/sub-device/disconnect

Method: POST

Produces: application/x-www-form-urlencoded,application/json

Consumes: */*

Note: To receive the push notification of the disconnection status of the add-ons.

Example:

```
{
  "deviceMac": "",
  "deviceName": "",
  "deviceType": 0,
  "onlineState": false,
  "radarId": "",
  "roomId": 0
}
```

Request parameter:

Parameter	Description	In	Required?	Data type	schema
subDeviceDTO	SubDeviceDTO	body	true	SubDeviceDTO	SubDeviceDTO
deviceMac	Sub device MAC address		false	string	
deviceName	Sub device name		false	string	
deviceType	Sub device type. Please refer to the enumeration value list for specific values		false	integer(int32)	
onlineState	Sub device online state. true: online false: offline		false	boolean	
radarId	radar id		true	string	
roomId	Assigned room id		false	integer(int64)	

Response status:

Status code	Description	schema
200	OK	restful api response object«Void»
201	Created	

Status code	Description	schema
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response parameter:

Parameter	Description	Type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.7. To push the firmware upgrade Status

url:/api/firmware/upgrade-status

method:POST

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: To push the firmware upgrade status

Example:

```
{
  "radarId": "",
  "radarVersion": "",
  "upgradeResult": false,
  "upgrading": false
}
```

Params:

name	description	in	require	type	schema
firmwareUpgradeStatusDTO	FirmwareUpgradeStatusDTO	body	true	FirmwareUpgradeStatusDTO	FirmwareUpgradeStatusDTO
radarId	Radar id		true	string	

name	description	in	require	type	schema
radarVersion	Current radar firmware version		true	string	
upgradeResult	Upgrade status of the radar firmware		true	boolean	
upgrading	Whether the radar upgrade will be repeatedly attempted within the 24hrs		true	boolean	

Status:

code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.8. Alert cancellation**url:**/api/eliminate-alarm**method:**POST**produces:**application/x-www-form-urlencoded,application/json**consumes:***/*

Note: This interface is applicable to the fall alert of the Assure and the motion alert of the PIR motion sensors.

Potential falls picked up by radar may not always generate a positive fall alert that is sent to the server, but may be cancelled if large movements are detected within the buffer time. In this case, the radar will report a fall-alert-cancellation to the server to cancel the previous alert that is sent via the /api/alarm port.

For the PIR motion sensors, whenever an alert is cancelled, the sub-device will report the alert cancellation with the MAC address of the device.

Example:

```
{
  "eliminateTime": "",
  "eliminateZonedTime": "",
  "functionId": 0,
  "radarId": "",
  "deviceMac":""
}
```

Params:

name	description	in	require	type	schema
eliminateAlarmDTO	EliminateAlarmDTO	body	true	EliminateAlarmDTO	EliminateAlarmDTO
eliminateTime	eliminateTime (yyyy-MM-dd HH:mm:ss)		true	string(date-time)	
eliminateZonedTime	ISO-8601 standard datetime		true	string(date-time)	
functionId	functionId, Please refer to the alert type enumeration value list for specific values		true	integer(int32)	
radarId	radar id		true	string	
deviceMac	Sub device MAC address		false	string	

Status:

code	description	schema
------	-------------	--------

code	description	schema
200	OK	restful api response object«Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.9. Battery life of the sub-device

url:/api/sub-device/power

method:POST

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: The battery life of the sub-device will be pushed when it is reported by the sub device.

Example:

```
{
  "voltage": 0,
  "power": 0,
  "deviceMac": "",
  "radarId": ""
}
```

Params:

name	description	in	require	type	schema
subDevicePowerD	SubDevicePower	body	true	SubDevicePowe	SubDevicePower

name	description	in	require	type	schema
TO	DTO			rDTO	DTO
voltage	Voltage Value		true	integer(int32)	
power	Battery Level Value		true	integer(int32)	
deviceMac	Sub-device MAC Address		true	string	
radarId	radar id		true	string	

Status:

code	description	schema
200	OK	restful api response object «Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.10. Multi-target detection and coordinates.**url:**/api/radar/person-position-detect**method:**POST**produces:**application/x-www-form-urlencoded,application/json**consumes:***/*

Note: The number and coordinates of the moving targets detected by the radar. It is refreshed every 10 s. It is available for radar firmware version of 2.8.1.44 and later. Please note that it reports moving targets only. Please note that the coordinates is for only one moving target. If there are more than one moving target, the coordinates will be the one that moves the earliest. If the targets starting moving in the same time, the coordinates will be the one nearest to the radar.

Example:

```
{
  "movingNum": 0,
  "motionSnr": 0,
  "motionXYZ": [],
  "personXYZ": [],
  "radarId": ""
}
```

Params:

name	description	in	require	type	schema
personPositionDetectionDTO	PersonPositionDetectionDTO	body	true	PersonPositionDetectionDTO	PersonPositionDetectionDTO
movingNum	Number of moving targets		true	integer(int32)	
motionSnr	Detection Signal Strength		true	number(float)	
motionXYZ	The coordinates (x,y,z) of the targets with minor motion		true	array	number(float)
personXYZ	The coordinates (x,y,z) of the moving target, for example a walker.		true	array	number(float)
radarId	radarId		true	string	

Status:

code	description	schema
200	OK	restful api response object «Void»
201	Created	
401	Unauthorized	

code	description	schema
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.11. Completion of the environmental interference threshold learning completion(Assure/Wavve/Wavve hub)

url:/api/room-interference-study-finished

method:POST

produces:application/x-www-form-urlencoded,application/json

consumes:*/*

Note: Once the device has completed learning of the environmental interference threshold, a completion notice will be pushed to the server via this interface. Please note this is applicable to threshold learning via API interface. Please refer to /api/v1/radar/save-room-interference-study-time/{radarId}) in the document “out cloud restful”.

Example:

```
{
  "radarId": ""
}
```

Params:

name	description	in	require	type	schema
roomInterferenceStudyFinishedDTO	RoomInterferenceStudyFinishedDTO	body	true	RoomInterferenceStudyFinishedDTO	RoomInterferenceStudyFinishedDTO
radarId	radarId		true	string	

Status:

code	description	schema
200	OK	restful api response object «Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

2.12. Real-time door sensor status push (Wavve hub)**url:**/api/door-state**method:**POST**produces:**application/x-www-form-urlencoded,application/json**consumes:***/***Note:** Real-time push of door open/close status..

- When the door is opened, the sensor will push : OPEN
- When the door is closed, the sensor will push: CLOSE

Example:

```
{
  "radarId": "",
  "roomId":0,
  "deviceMac":"","
  "doorState":"CLOSE"
}
```

Params:

name	description	in	require	type	schema
doorInfoDTO	DoorInfoDTO	body	true	DoorInfoDTO	DoorInfoDTO
radarId	radarId		true	string	
roomId	roomId		true	integer(int64)	
deviceMac	Sub-device MAC Address		true	string	
doorState	doorState		true	string	

Status:

code	description	schema
200	OK	restful api response object «Void»
201	Created	
401	Unauthorized	
403	Forbidden	
404	Not Found	

Response Params:

name	description	type	schema
code	response code	string	
msg	response description message	string	

Response Example:

```
{
  "code": "200",
  "msg": "success"
}
```

3. Support

Please contact support@ax-end.com for any help or more information.

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